

Final Project — Part I

Numerical Methods for Ordinary Differential Equations

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1 Introduction

In this project I compare several numerical methods for solving the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha.$$

The methods I implemented are Euler's method, the second-order Taylor method, the midpoint method, the modified Euler method, Heun's third-order method, the classical fourth-order Runge–Kutta method (RK4), the four-step Adams–Bashforth method, and the fourth-order Adams predictor–corrector method (whose corrector is the three-step Adams–Moulton formula). My goals are to compare their accuracy, to see how the error behaves as the step size h shrinks, to check whether the observed error matches the theoretical order of each method, and to look at how error propagates and how stability differs from one method to another.

All methods were coded in Python; the code is described in Section 7. As a check, my implementations reproduce the textbook values for Problem A to every printed digit (for example, the midpoint and Heun columns of Tables 5.6–5.7 in the notes).

2 The Methods

Let $t_i = a + ih$ and let $w_i \approx y(t_i)$. The one-step methods are

$$\text{Euler: } w_{i+1} = w_i + hf(t_i, w_i),$$

$$\text{Taylor (order 2): } w_{i+1} = w_i + hf + \frac{h^2}{2} f', \quad f' = f_t + f_y f,$$

$$\text{Midpoint: } w_{i+1} = w_i + hf\left(t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)\right),$$

$$\text{Modified Euler: } w_{i+1} = w_i + \frac{h}{2} [f(t_i, w_i) + f(t_{i+1}, w_i + hf(t_i, w_i))],$$

$$\text{Heun (order 3): } w_{i+1} = w_i + \frac{h}{4} \left[f(t_i, w_i) + 3f\left(t_i + \frac{2h}{3}, w_i + \frac{2h}{3}f\left(t_i + \frac{h}{3}, w_i + \frac{h}{3}f(t_i, w_i)\right)\right) \right],$$

and RK4 uses the usual four slope estimates $k_1 = hf(t_i, w_i)$, $k_2 = hf(t_i + \frac{h}{2}, w_i + \frac{k_1}{2})$, $k_3 = hf(t_i + \frac{h}{2}, w_i + k_2)$, $k_4 = hf(t_{i+1}, w_i + k_3)$,

$$w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

The multistep methods reuse previously computed slopes. The explicit four-step Adams–Bashforth method is

$$w_{i+1} = w_i + \frac{h}{24} (55f_i - 59f_{i-1} + 37f_{i-2} - 9f_{i-3}),$$

and the fourth-order Adams predictor–corrector method uses this formula to predict w_{i+1}^* , then corrects once with the implicit three-step Adams–Moulton formula

$$w_{i+1} = w_i + \frac{h}{24} (9f(t_{i+1}, w_{i+1}^*) + 19f_i - 5f_{i-1} + f_{i-2}).$$

Because the multistep methods need the starting values w_1, w_2, w_3 , I generate those with RK4 before switching over, as in the textbook algorithm. The nominal global orders are: Euler $O(h)$; Taylor, midpoint, and modified Euler $O(h^2)$; Heun $O(h^3)$; and RK4, Adams–Bashforth, and the Adams predictor–corrector $O(h^4)$.

3 Problem A

Consider

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

with exact solution $y(t) = (t+1)^2 - \frac{1}{2}e^t$. Table 1 gives an overview: the maximum error of all eight methods at $h = 0.1$, together with the observed order. Every method behaves at its nominal order, and within an order group the leading constant still matters — for instance the midpoint method is about five times more accurate than the modified Euler method even though both are order 2.

Table 1: Problem A: maximum absolute error and observed order for all eight methods at $h = 0.1$.

Method	max error ($h = 0.1$)	observed order	nominal order
Euler	2.42e-01	0.86	1
Taylor (order 2)	1.14e-02	1.89	2
Midpoint	3.75e-03	2.01	2
Modified Euler	1.89e-02	1.94	2
Heun (order 3)	5.73e-05	3.06	3
RK4	6.99e-06	3.96	4
Adams–Bashforth 4	1.85e-04	3.46	4
Adams PC 4	1.09e-05	3.21	4

Table 2 then lists the absolute error of five representative methods at each grid point for $h = 0.2$, and Figure 1 shows the solution and the errors.

Table 2: Problem A: absolute error at each grid point, $h = 0.2$.

t_i	$y(t_i)$	Euler	Midpoint	Heun	RK4	Adams PC
0.0	0.5000000	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
0.2	0.8292986	2.93e-02	1.30e-03	5.42e-05	5.29e-06	5.29e-06
0.4	1.2140877	6.21e-02	2.73e-03	1.13e-04	1.14e-05	1.14e-05
0.6	1.6489406	9.85e-02	4.28e-03	1.75e-04	1.86e-05	1.86e-05
0.8	2.1272295	1.39e-01	5.95e-03	2.39e-04	2.69e-05	2.39e-05
1.0	2.6408591	1.83e-01	7.69e-03	3.04e-04	3.64e-05	3.05e-05
1.2	3.1799415	2.30e-01	9.48e-03	3.65e-04	4.74e-05	3.89e-05
1.4	3.7324000	2.81e-01	1.12e-02	4.20e-04	5.99e-05	4.95e-05
1.6	4.2834838	3.33e-01	1.29e-02	4.61e-04	7.43e-05	6.30e-05
1.8	4.8151763	3.87e-01	1.42e-02	4.80e-04	9.06e-05	7.99e-05
2.0	5.3054720	4.40e-01	1.51e-02	4.65e-04	1.09e-04	1.01e-04

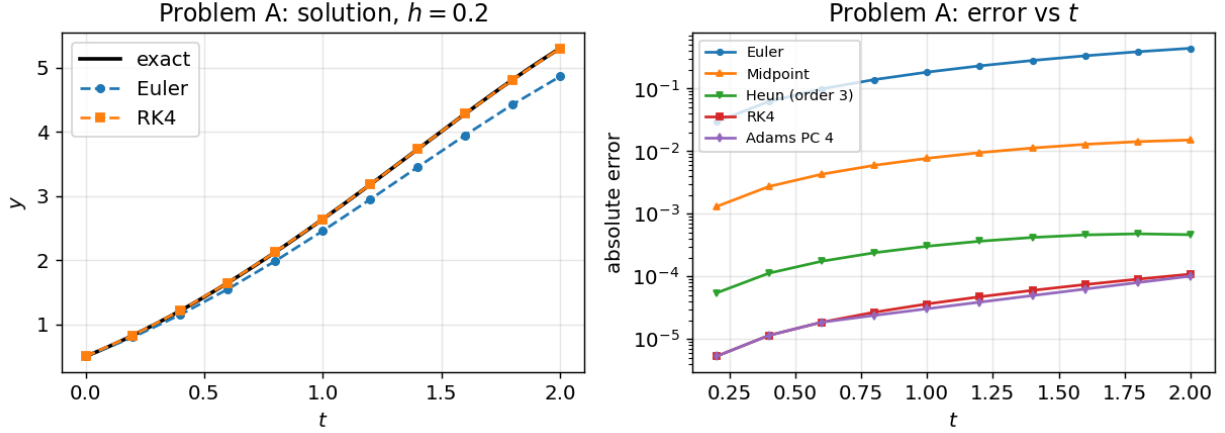


Figure 1: Problem A. Left: Euler lags the exact curve while RK4 sits on top of it. Right: the errors separate cleanly by order — Euler ($\sim 10^{-1}$), midpoint ($\sim 10^{-2}$), Heun ($\sim 10^{-4}$), and RK4 $\approx$ Adams PC ($\sim 10^{-5}$).

Euler is by far the least accurate: its error at $t = 2$ is about 0.44 , while RK4 and the Adams predictor–corrector are near 10^{-4} , an improvement of roughly four orders of magnitude for the same step size. To check that these methods actually achieve their advertised order, I recomputed the maximum error as h was halved repeatedly; Table 3 reports the error and the observed order $\log_2(e_{2h}/e_h)$.

Table 3: Problem A: maximum absolute error and observed order (in parentheses) as h is halved.

h	Euler	Midpoint	Heun	RK4	Adams PC
0.2000	4.40e-01 (–)	1.51e-02 (–)	4.80e-04 (–)	1.09e-04 (–)	1.01e-04 (–)
0.1000	2.42e-01 (0.86)	3.75e-03 (2.01)	5.73e-05 (3.06)	6.99e-06 (3.96)	1.09e-05 (3.21)
0.0500	1.27e-01 (0.92)	9.28e-04 (2.01)	7.01e-06 (3.03)	4.42e-07 (3.98)	9.13e-07 (3.58)
0.0250	6.55e-02 (0.96)	2.30e-04 (2.01)	8.66e-07 (3.02)	2.78e-08 (3.99)	6.60e-08 (3.79)
0.0125	3.32e-02 (0.98)	5.74e-05 (2.01)	1.08e-07 (3.01)	1.74e-09 (4.00)	4.43e-09 (3.90)

The observed orders settle down to 1, 2, 3, 4 for Euler, midpoint, Heun, and RK4 respectively, matching the theoretical local-truncation-error analysis. The Adams predictor–corrector approaches order 4 as well; it starts a little below 4 on the coarsest grids because of the RK4 start-up values and then improves as h shrinks.

4 Problem B

Consider

$$y' = 2y, \quad 0 \leq t \leq 2, \quad y(0) = 1,$$

with exact solution $y(t) = e^{2t}$. This solution grows exponentially, which makes it a good test of how error propagates. Table 4 compares Euler and RK4 at $h = 0.1$.

Table 4: Problem B: exponential growth, $h = 0.1$.

t_i	$y(t_i)$	Euler	Euler error	RK4 error
0.0	1.000000	1.000000	0.000000	0.00e+00
0.2	1.491825	1.440000	0.051825	6.74e-06
0.4	2.225541	2.073600	0.151941	2.01e-05
0.6	3.320117	2.985984	0.334133	4.50e-05
0.8	4.953032	4.299817	0.653215	8.95e-05
1.0	7.389056	6.191736	1.197320	1.67e-04
1.2	11.023176	8.916100	2.107076	2.99e-04
1.4	16.444647	12.839185	3.605462	5.20e-04
1.6	24.532530	18.488426	6.044104	8.86e-04
1.8	36.598234	26.623333	9.974901	1.49e-03
2.0	54.598150	38.337600	16.260550	2.47e-03

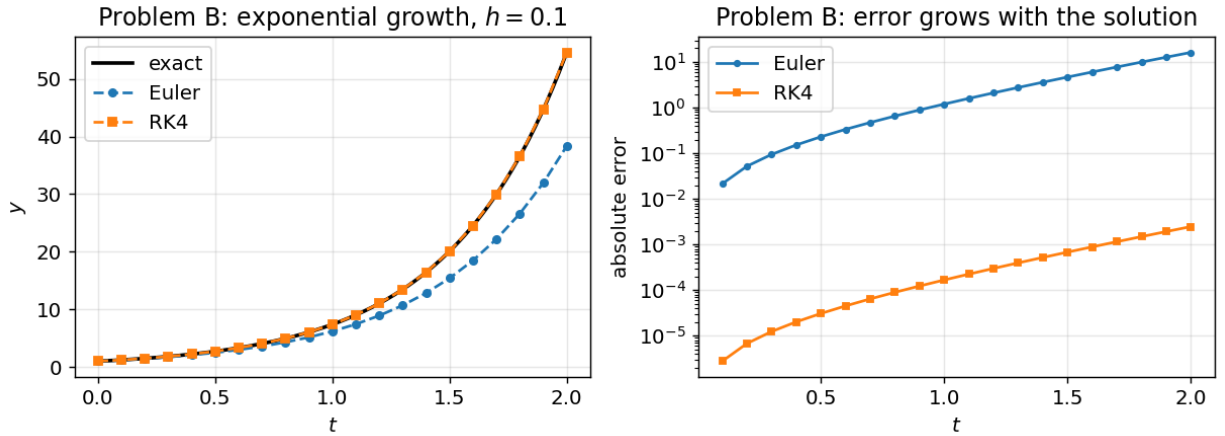


Figure 2: Problem B. The Euler error grows right along with the solution, while RK4 stays accurate. Note the logarithmic scale on the right.

The Euler error grows from about 0.05 at $t = 0.2$ to over 16 at $t = 2$. This is exactly what error propagation predicts: when $y' = 2y$, the factor $f_y = 2 > 0$ amplifies any existing error at every step, so the global error itself grows roughly like e^{2t} . RK4 keeps the error near 10^{-3} over the whole interval — a higher-order method is much better at keeping the propagated error small, but the underlying growth of the problem is still visible in the slow climb of the RK4 error curve.

5 Problem C (student-designed)

For my own problem I chose

$$y' = -10y + \sin t, \quad 0 \leq t \leq 3, \quad y(0) = 1,$$

with exact solution

$$y(t) = \frac{10 \sin t - \cos t}{101} + \frac{102}{101} e^{-10t}.$$

I picked this because the $-10y$ term causes a rapid initial decay (a mild “stiffness”), which is exactly the situation where explicit methods can run into trouble if the step size is too large. Table 5 shows the maximum error of each method at the stable step $h = 0.1$.

Table 5: Problem C: maximum absolute error at $h = 0.1$.

Method	max error ($h = 0.1$)	order
Euler	3.72e-01	1
Midpoint	1.33e-01	2
Heun	3.49e-02	3
RK4	7.19e-03	4
Adams PC	2.42e-02	4

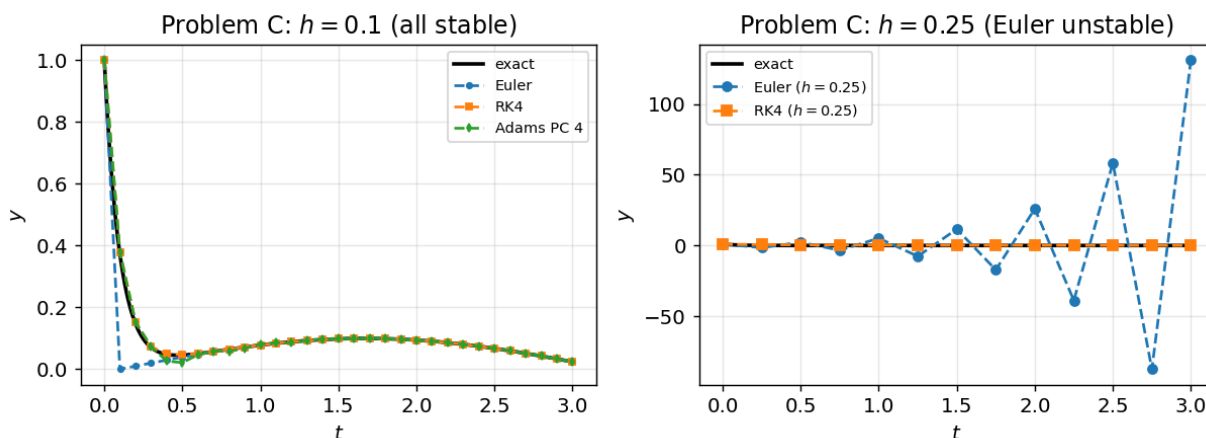


Figure 3: Problem C. Left: at $h = 0.1$ every method tracks the exact solution (Euler is slightly rough through the sharp initial decay). Right: at $h = 0.25$ Euler is outside its stability region and oscillates with growing amplitude, while RK4 — which has a larger stability region — stays accurate.

Even at the stable step the errors here are larger than in Problem A, because the steep transient near $t = 0$ is hard to resolve; the largest error for every method occurs in that region. The stability difference shows up clearly when I enlarge the step. For Euler, stability on this problem requires $-2 < h\lambda < 0$ with $\lambda = -10$, i.e. $h < 0.2$. At $h = 0.25$ this is violated: Euler oscillates and blows up to $w(3) \approx 131$ even though the true solution is about 0.024. RK4 has a larger real-axis stability region and remains stable at $h = 0.25$ ($w(3) \approx 0.028$). Pushing further to $h = 0.3$ eventually destabilizes RK4 too, which confirms that explicit methods trade a larger stability region for more work per step but never have an unbounded one.

6 Discussion

Which methods are most accurate? For a fixed step size, accuracy tracks the order of the method. RK4 and the Adams predictor–corrector (both order 4) were the most accurate in every experiment, followed by Heun (order 3), the order-2 methods, and finally Euler.

How does decreasing h affect the error? Each time h is halved, the error drops by a factor of about 2^p , where p is the order (Table 3): roughly $2\times$ for Euler, $4\times$ for the order-2 methods, $8\times$ for Heun, and $16\times$ for RK4.

Does the observed error agree with the theory? Yes. The measured orders in Table 3 converge to the predicted 1, 2, 3, 4, which is the clearest confirmation that the implementations are correct and that the truncation-error analysis describes the real behavior.

Why does RK4 usually beat Euler? Euler uses only the slope at the left endpoint, so it commits an $O(h^2)$ error every step. RK4 samples the slope four times per step and combines them so that error terms through h^4 cancel, giving local error $O(h^5)$ and global error $O(h^4)$. It costs four function evaluations per step instead of one, but the accuracy gain is enormous.

Why can methods of the same order behave differently? The midpoint and modified Euler methods are both order 2, yet on Problem A the midpoint method was about five times more accurate (Table 1). They have the same order but different leading error constants, so “same order” only means the errors shrink at the same *rate*, not that they are the same size. The same is true of RK4 versus the Adams predictor–corrector: both order 4, but with slightly different constants.

How does error propagate over time? On Problem A and especially Problem B the error grows as t increases, because each step’s new error is carried forward and, when $f_y > 0$, amplified. On Problem C, where $f_y = -10 < 0$, the exact dynamics instead damp errors — as long as the method is stable.

Which methods appear more stable? For a fixed step size the higher-order Runge–Kutta methods have larger stability regions, so RK4 tolerated a step ($h = 0.25$) on Problem C that made Euler blow up. But no explicit method is unconditionally stable: a large enough step destabilizes all of them.

7 Code

All methods are implemented in `ode_methods.py`, and `run_project.py` reproduces every figure and table in this report; `check_regression.py` verifies the output against the published textbook values. The code is available at:

<https://github.com/Hovatar/final-projects/tree/master/math104c-final-project-1>